



Recent advances in object tracking using hyperspectral videos: a survey

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Abstract

Short-Term Single-Object (STSO) tracking using Hyperspectral Videos (HSVs), which has become a hotspot recently, is a challenging task. Hyperspectral Object Tracking (HOT) makes full use of spatial and spectral information during the tracking process. In HOT, multiple features, including deep network features, have been combined with correlation filter methods, which increases time-consuming efficiency and tracking accuracy. However, redundant spectral information needs to be obtained in an effective way. In addition, there is currently no detailed investigation of HOT algorithms. Therefore, this survey studies the development of HOT algorithms in recent years. Specifically, several HSVs are listed, an investigation of HOT algorithms is conducted, and components of HOT are described in detail. Furthermore, several popular HOT algorithms, including our previous work BS-SiamPRN and AD-SiamRPN, are compared quantitatively and qualitatively. Finally, the research status of HOT is summarized, and future work has been described, which lays the foundation for future HOT or STSO referring to HSVs.

Keywords Hyperspectral video · Target tracking · Correlation filter · Deep learning

1 Introduction

Object tracking has been a popular area of research in computer vision for several decades, such as face tracking [5], autonomous driving [76, 86], aerial tracking [11, 41, 87], environmental protection [25, 46, 51] and security monitoring [14, 48, 58]. However, complex scenarios, including occlusions, background clutter, fast motion, low sampling rate [56], and object deformation, have made target tracking using RGB videos a challenging area of research. Especially, when the appearance of a target is similar to that of the background, these visual trackers tend to lose targets. To solve these problems, some researchers concen-

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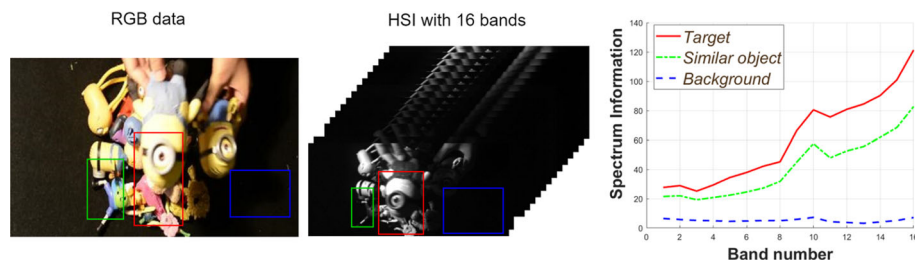


Fig. 1 The two toys (a plastic one and a plush one) with the same color, have different spectral curves

trate on Hyperspectral Videos (HSVs) processing [1, 8, 16, 26, 52], which may originate 30 years ago. The hyperspectral data cube contains spatial-spectral information [39], which is shown in Fig. 1. The Hyperspectral Image (HSI) has been applied in environmental monitoring [35], aerial object tracking [70], geological mapping [21], space object detection [74, 96] and biological applications [2]. Moreover, there have been some studies and methods aimed at developing object tracking algorithms for Hyperspectral images (HSIs).

In the field of tracking, there are two main methods, including Discriminative Correlation Filter (DCF) based methods [4, 17, 19, 23, 28, 36, 44, 97] and Deep Learning (DL) based methods [27, 29, 30, 50, 79, 81, 83]. However, the RGB feature can not effectively deal with similar color between the target and its neighborhood, which is not a problem using spectral information in HSVs. So far, some researchers have done well in Hyperspectral Object Tracking (HOT). However, there is a lack of a review that gives a more complete picture of research in this area. Therefore, in this paper, a comprehensive survey of HOT is given, whose structure is described in Fig. 2.

The contributions of this paper are summarized below.

- The object tracking methods on hyperspectral videos are discussed, and categories are subdivided on the basis of their methodologies.

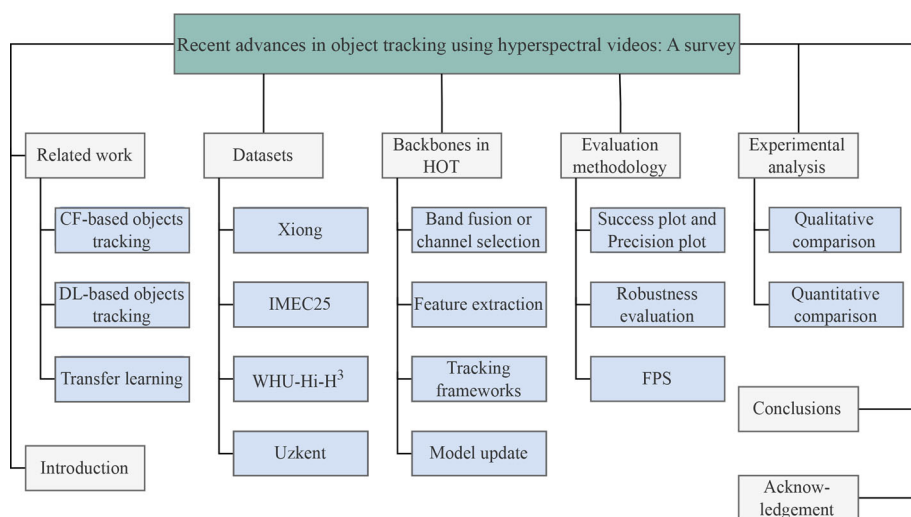


Fig. 2 The structure of this paper

- A comprehensive survey is given, including data sets, tracking methods, and a summary of experimental results.
- We analyze the result of hyperspectral tracking and give a comparison to some popular methods, containing our previous method.
- Furthermore, some conclusions and possible research directions for hyperspectral video tracking are discussed.
- We conclude that the Hyperspectral Object Tracking (HOT) referring to the Siamese network achieves a high precision and real-time performance. In addition, deep learning methods that train raw spectral data are more effective than tracking methods based on band selection. Furthermore, transfer learning technology helps fuse semantic information into the DCF tracker with large data sets.

2 Related work

This section highlights two prevalent techniques for target tracking, including CF-based object tracking and DL-based object tracking, while introducing the relevant concept of transfer learning. Subsequently, the following section underscores the subject matter of target tracking based on hyperspectral video.

2.1 CF-based objects tracking

The DCF trains filters [31, 47] efficiently with Fast Fourier Transform (FFT). DCF can learn a correlation filter that identifies the object and its neighborhood. The fast Minimum Output Sum of Squared Error (MOSSE) tracker [4] extracts features from grayscale data and reaches a faster speed. However, MOSSE faces difficulties in the complex environment because only the raw gray part is utilized for tracking with fewer semantic information. Moreover, some DCF trackers use other features, including Histogram of Gradient (HOG) features [17], colornames features [12], and DL features [50]. The Kernelized Correlation Filter (KCF) [17] makes the best use of HOG features. In addition, the kernel trick is applied to KCF which has a high efficiency. The HOG-based feature extraction method plays a crucial role in accurately determining the edges of the target and improving the tracking accuracy of objects with significant contours. Furthermore, the Spatially Regularized Discriminative Correlation Filters (SRDCF) [10] can effectively reduce the boundary effect and eliminate the target area. The Multi-Cue Correlation based Tracker (MCCT) [82] uses HOG features, color names feature, and VGG (Visual Geometry Group), which represents the inherent attributes of the target. These DCF methods deal with the occlusion challenge [22] of the original MOSSE.

2.2 DL-based objects tracking

Trackers based on deep learning focus on deep features extraction, such sparse autoencoder based methods [81], VGG based methods [71] and siamese-based network methods [7, 29, 72, 75]. Most of them use offline training and end-to-end online tracking to update parameters and extract the inherent characteristics of the object. The Deep Learning Tracker (DLT) tracker [81] is the first to apply deep network to track, but the feature is not excellent in describing the target. The middle layers of the VGG network [79] can be utilized to obtain strong features referring to targets. However, this network has a fixed input size.

Siamese-based trackers are commonly applied in object tracking and can be split into anchor-based and anchor-free trackers. The first time you apply the Siamese network to tracking tasks is the Siamese INstance search for Tracking (SINT) [65], which utilizes two inputs fed into two a same network, and it is also the first time that the target tracking is transformed into a patch block matching problem. The Siamese Fully Convolutional (SiamFC) [7] achieves great results with an amount of offline data for training, and the algorithm speed reaches 85 Frames Per Second (FPS). Dsiam [13] applies a dynamic siamese approach by two online transformations, obtaining better tracking accuracy under the premise of high speed. The Siamese Region Proposal Network (SiamRPN) [42] revolutionizes the Siamese tracking approach by incorporating a regional proposal network to handle scale variations. The extracted feature is divided into a classification branch and a regression branch. Specifically, the classification branch plays a crucial role in differentiating the foreground and background regions of an image. In addition, the regression branch selects the appropriate intersection on the union and fine-tunes the final candidate box for the predicted output target. The latter can ignore the original template scale and improve the performance of scale variation. In essence, the SiamRPN method transforms the initial comparison between the target and surrounding regions into a binary classification problem and a regression problem for localization. The features extracted from a siamese network are similar to the framework of SiamFC [7].

The classification branch and the regression branch use features extracted by a Siamese network to manipulate template frames and detection frames, respectively, and carry out convolution operations. The relevant equations are as follows.

$$T_{w \times h \times 2k}^{cls} = \psi(x)_{cls} * \psi(z)_{cls} \quad (1)$$

$$T_{w \times h \times 4k}^{reg} = \psi(x)_{reg} * \psi(z)_{reg} \quad (2)$$

where w and h show the size of target candidate boxes. x means the template frame and z means the search frame. $\psi(x)$ and $\psi(z)$ represent the output features of the detection branch and the template branch in the Siamese network, respectively. The $T_{w \times h \times 2k}^{cls}$ tensor holds $2k$ channels, with each channel corresponding to the positive and negative activation of the target. The classification loss is computed by using the softmax function on this tensor. $T_{w \times h \times 4k}^{reg}$ contain $4k$ channels. In summary, the definition of the loss function for the regression branch is provided in (3). Equation (4) shows some details about it.

$$L_{reg} = \sum_{i=1}^4 smooth_{L_r}(\delta[i], \sigma) \quad (3)$$

$$\begin{aligned} \delta[1] &= (G_x - C_x)/C_w \\ \delta[2] &= (G_y - C_y)/C_h \\ \delta[3] &= \ln(G_w/C_w) \\ \delta[4] &= \ln(G_h/C_h) \end{aligned} \quad (4)$$

where the anchor frame's center point coordinates, width, and height are represented by C_x , C_y , C_w , and C_h , while the ground truth's center point coordinates and dimensions are represented by G_x , G_y , G_w , and G_h . The $smooth_{L_r}$ loss, as defined in (3), is calculated as a measure of smoothness. σ is a parameter used to control the smoothness or shape of the smooth L1 loss function, which is commonly used in regression tasks.

$$smooth_{L_r} = \begin{cases} 0.5\sigma^2 x^2 & \text{if } |x| < 1/\sigma^2 \\ |x| - 1/2\sigma^2 & \text{if } |x| \geq 1/\sigma^2 \end{cases} \quad (5)$$

Then the total loss function L_{tol} is

$$L_{tol} = L_{cls} + \lambda L_{reg} \quad (6)$$

where the hyperparameter λ balances the regression loss and classification loss, and the cross-entropy loss is represented by L_{cls} . The final refined top proposal set can be output as

$$\begin{aligned} x^{pro} &= x^{an} + d_x^{reg} \times w^{an} \\ y^{pro} &= y^{an} + d_y^{reg} \times h^{an} \\ w^{pro} &= w^{an} \times w^{d_w} \\ h^{pro} &= h^{an} \times h^{d_h} \end{aligned} \quad (7)$$

where the superscript word *pro* means the proposal value, *an* denotes the anchor point value; and *reg* means the value in the regression branch.

There are some improved trackers based on SiamRPN, such as DaSiamRPN [98] and SiamRPN++ [15]. The DaSiamRPN [98] makes use of data enhancement for improving the tracking performance. The SiamRPN++ [15] uses ResNet-50 as its backbone for using the deep network to extract features, whose speed reaches over 120 fps. In addition, the anchor-free trackers, such as Ocean [43] and SiamCAR [94], do not require anchors without designing the anchors' parameters.

2.3 Transfer learning

In the hyperspectral video tracking community, Transfer Learning (TL) focuses on learning knowledge in the RGB data source domain R_s and the learning task H_s , improving the learning of prediction function $f(\cdot)$ in the hyperspectral video data target domain R_t .

$$f(x_t) = \langle \omega(\chi), \varphi(x_s) \rangle \quad (8)$$

where $x_t \in R_t$, $x_s \in R_s$, $\chi = \{x_t, x_s\}$, ω denotes the shared parameters for target task H_t and φ denotes the model parameter for H_s .

Deep learning models are usually trained for single object tracking based on large datasets. However, with small datasets, it is necessary to first perform pretraining on a large dataset for other tasks. Then, the trained model can be fine-tuned using the specific, smaller target dataset while sharing the weight. HSVs possess a wealth of spectral and spatial information, which can partially compensate for the deficiencies in RGB data. However, the DL model does not perform well in target tracking using only the current HSV. Therefore, the transfer learning strategy aims to enhance the classification performance of features by leveraging information from other tasks and data sets. Specifically, it tends to use RGB data to improve the process of target tasks for pre-training, such as ImageNet [55], COCO [34], GOT-10K [20].

From the model perspective, the TL technique can be implemented by sharing certain parameters between the model trained on the RGB data source domain and the hyperspectral video model, allowing for improving the performance in the target domain. The pre-trained model using the RGB data source domain makes the trackers stronger [32], and reaches a better generalization ability. Nevertheless, pre-training with RGB data results in some loss of semantic information of hyperspectral data. Thus, features with strong classification performance need to be obtained.

Recently, several HSV tracking algorithms have been proposed, which are listed in Table 1, where the words marked with * mean that we create the abbreviations.

Table 1 Examples of published research on HOT

Reference	Years	Algorithm	Journal/Conference
Banerjee et al. [3]	2009	IIT*	WHISPERS
Uzkent et al. [67]	2015	SVM*	Procedia Computer Science
Uzkent et al. [68]	2016	HFT	CVPR Workshops
Uzkent et al. [69]	2017	HLT	CVPR Workshops
Uzkent et al. [70]	2018	DeepHKCF	IEEE Transactions on Geoscience and Remote Sensing
Qian et al. [54]	2018	CNHT	International Conference on Smart Multimedia
Xiong et al. [85]	2020	MHT	IEEE Transactions on Image Processing
Li et al. [40]	2020	BAE-Net	ICIP
Liu et al. [37]	2021	HA-Net	WHISPERS
Li et al. [43]	2021	SST-Net	WHISPERS
Zhang et al. [95]	2021	MFI-HVT	WHISPERS
Liu et al. [45]	2021	H ³ Net	IEEE Transactions on Geoscience and Remote Sensing
Chen et al. [9]	2021	SS_ECO	Remote Sensing
Zhao et al. [92]	2022	RA-Net	Remote Sensing
Lei et al. [33]	2022	SSDT-Net	Remote Sensing
Zhao et al. [91]	2022	DC-HVT	Remote Sensing
Zhang et al. [100]	2022	SMR-AS3D*	Remote Sensing
Zhang et al. [93]	2022	FCSS*	WHISPERS
Su et al. [57]	2022	TrTSN	WHISPERS
Wang et al. [80]	2022	BS-SiamRPN	WHISPERS
Wang et al. [78]	2022	SSATFN	WHISPERS
Liu et al. [38]	2022	SiamHYPER	IEEE Transactions on Image Processing
Hou et al. [18]	2022	TSCFW	IEEE Transactions on Geoscience and Remote Sensing
Qian et al. [53]	2023	GOMT	IET Image Processing

3 Datasets

As far as we know, there are four Hyperspectral Video (HSV) datasets, three of which are real scene datasets [9, 45, 85], and another is a simulation dataset [70]. Some details are shown in Table 2.

3.1 Xiong dataset

Xiong's dataset [85] consists of 40 training videos and 35 testing videos published in 2020 on hsitracking.com, each including RGB videos, HSVs, and false-color videos. At present, there have been alterations made to the training and validation datasets, entailing an augmentation of 15 training videos and 11 testing videos, covering 16 spectral bands. Additionally, the integration of two additional XIMEA snapshot cameras, specifically the NIR and Red-NIR cameras, has expanded the spectral coverage to 25 bands (comprising 40 training and 30 validation datasets) and 15 bands (encompassing 15 training and 11 validation datasets), respectively, after a meticulous calibration procedure. Some RGB images are shown in Fig. 3. These videos were collected using a snapshot mosaic hyperspectral camera called the XIMEA

Table 2 Dataset comparison

Dateset	Name			
	Xiong	IMEC25	WHU-Hi-H ³	Uzkent
Attribute	Real	Real	Real	Similar
Years	2020	2021	2021	2018
Public	Yes	No	No	Yes
Pixels	512×256	409×216	409×216	184×200
channels	16,25,15	25	25	6
band	470~620	680~960	600~900	400~1000
range /nm				
Challenge	SV,OPR,IPR BC,OCC,IV DEF,MB FM,OV,LR	SV,OPR,IPR BC,OCC,IV DEF,MB FM,OV,LR	IV,DEF,SV BC,LR,FM	Details in [70]
Scenes	Total 197 videos details in [85] https://hsitracking.com	Total 135 videos navigation, aviation, traffic	Cars, coin bracelet,bag pedestrians persons park, shadow green box	Vehicle
Each sequence	About 425 frames	Average 174 frames	101 frames	Details in [70]
Exist false-color videos	Yes	Yes	No	No

SSM4x4 VIS camera. Each sequence has an average of about 425 frames, and each frame consists of a 3-dimensional hyperspectral cube that comprises 512×256 pixels and 16 frequency bands ranging from 470nm to 620nm. The common challenge of this dataset contains

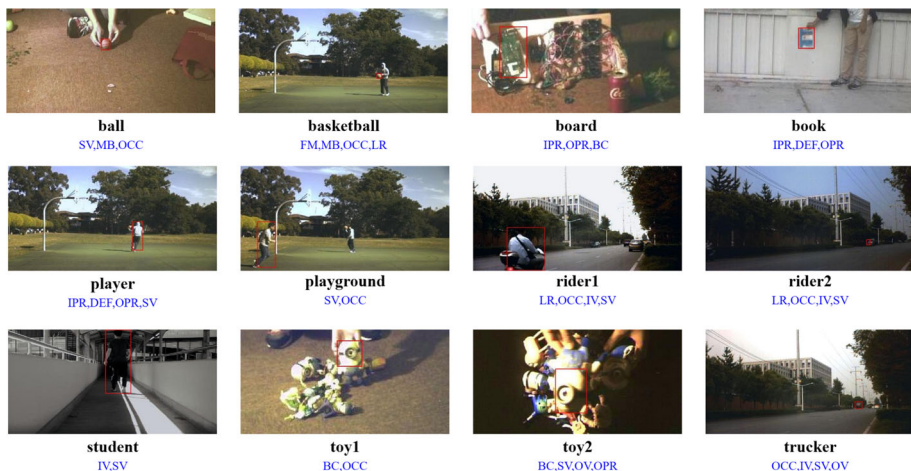
**Fig. 3** Some images in Xiong's dataset

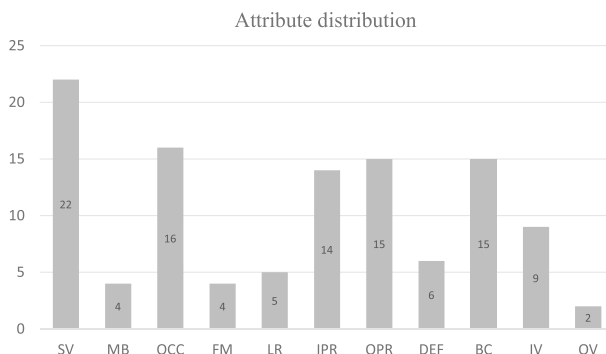
Table 3 Attributes description

Attributes	Description
IV	The illumination in the tracking region is changed significantly.
SV	The current frame is outside the range $\left[\frac{1}{\xi}, \xi\right], \xi > 1$
OCC	The object is occluded partially by other distractors or background.
DEF	The target deforms non-rigid objects in the scene.
FM	The motion of the target is larger than t_s pixels ($t_s = 20$)
IPR	The target rotates in the image plane.
OPR	The target rotates out of the image plane.
OV	The target leaves the view partly or fully.
MB	Due to the motion of the target or camera, the object is blurred.
BC	The background is similar to the target in color or texture.
LR	The number of pixels in the bounding box is less than a specific value (commonly, the value=400)

Scale Variation (SV), Out-of-Plane Rotation (OPR), In-Plane Rotation (IPR), Background Clutters (BC), OCCLUSION (OCC), Illumination Variation (IV), DEFORMATION (DEF), Motion Blur (MB), Fast Motion (FM), Out-of-View (OV), and Low Resolution (LR), which are also difficulties for RGB tracking. We introduce an approach to characterize the sequences by labeling them with the 11 attributes depicted in Table 3, providing a more comprehensive understanding of the sequences. In addition, Fig. 4 shows the distribution of each attribute.

3.2 IMEC25 dataset

Chen et al. [9] use a snapshot mosaic hyperspectral camera, named CMV2K SSM5×5 VIS. The dataset contains 80 training videos and 55 testing videos, called IMEC25, which is on <https://github.com/Chenlulu1993/HOMGIMEC25> Dataset. IMEC25 has 25 bands ranging from 680nm to 960nm, which belongs to Near Infrared (NIR) band. Its spatial resolution is 409×216 pixels for each band. Besides, the average length of all videos is 174 frames. The

**Fig. 4** The distribution of attributes in Xiong's dataset

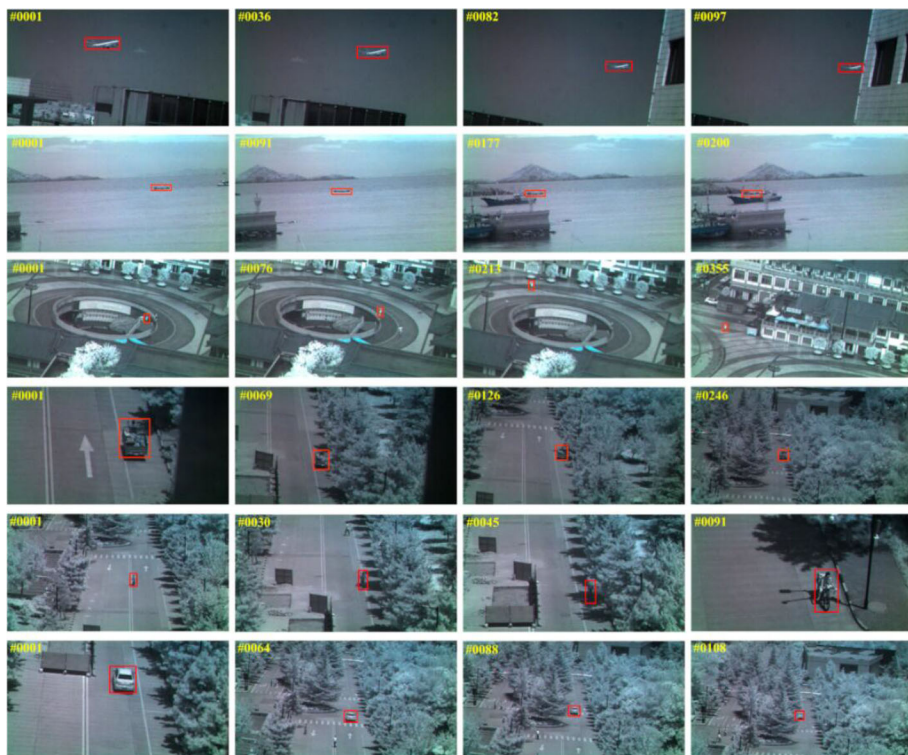


Fig. 5 Some example sequence of the IMEC25 dataset

dataset captured three real-world scenarios, including navigation, aviation, and traffic. Similar to the operation of the first dataset, it also contains false-color videos, which used three spectral bands (870.95nm, 808.64nm, 747.12nm) in order to compare color-based trackers and hyperspectral-based trackers. This sequence has 11 challenging attributes, including IV, SV, IPR, OPR, OV, BC, OCC, DEF, MB, FM, and LR. Fig. 5 shows some videos sequence in the IMEC25 dataset.

3.3 WHU-Hi-H³ datasets

The WHU-Hi-H³ datasets [45] were captured by a 25-channel snapshot spectral camera and contain 409×216 pixels and 25 bands ranging from 600nm to 900nm with a total of 6969 frames. Liu et al. designed nine camouflage target scenes, including cars, coins, bracelets, bags, pedestrians, persons, parks, green boxes, and shadows. Besides, the dataset is composed of nine scenes separated into 69 sequences, where the per sequence has 101 frames. The dataset is also labeled with the same 11 challenging attributes, some images of the WHU-Hi-H³ dataset are shown in Fig. 6.



Fig. 6 Some images in the WHU-Hi-H³ dataset

3.4 Uz Kent dataset

Uz Kent et al. [69] collect 30 channels in the visible wavelength range at 2Hz overlooking a university's parking. The dataset consists of 184×200 pixels, but it may not be public. However, Uz Kent [70] creates a synthetic dataset for single-channel vehicle classification using DIRSIG (Digital Imaging and Remote Sensing) software, and then fine-tuned a neural network for application on the Wide Area Motion Imagery platform.

The dataset includes nine different scenes, each set to 0.2 fps and producing 20 images to generate the RITMOS (Rochester Institute of Technology Multi-object Spectrometer)-like scenario with a total of 61 channels. The bands roughly range from 400nm to 1000nm, with six sampled bands. The dataset includes 55226 chips captured from mega-scene. The dataset provides the contribution for vehicle detection from aerial platforms, namely WAMI (Wide Area Motion Imagery).

4 Tracking process in HOT

Compared to color-based tracking systems, hyperspectral trackers generally entail a greater number of operations, such as feature dimensionality reduction. The present section endeavors to partition the tracking processing into four discrete components predicated on commonly employed tracking methodologies: band fusion or channel selection, feature extraction, tracking frameworks, and model update.

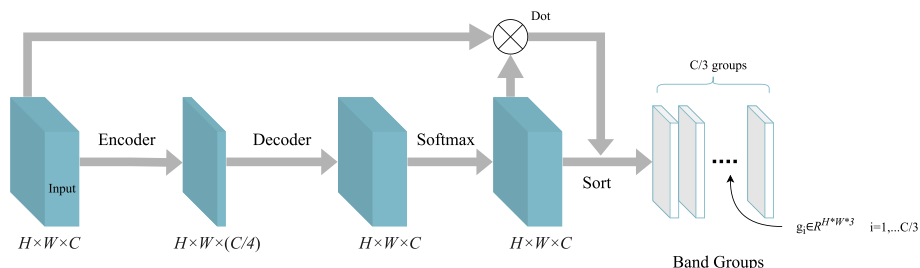


Fig. 7 The band attention module in BAE-Net [40]

4.1 Band fusion or channel selection

There is much information included in a hyperspectral image, which is time-consuming for object tracking. To streamline the tracking process and eliminate excessive duplication in HOT, feature dimension reduction techniques should be employed, such as feature selection [84] and feature extraction [99]. Conversely, to optimize the efficacy of extant color-based tracking systems, feature dimensionality reduction operations are imperative. Uz Kent et al. [69] propose a grouping-based approach using the same specific group wavelength channels and representing every five bands in a single likelihood map, a total of 12 maps. Two widely used techniques for band fusion are the variance ratio method [88] and the sum-rule method [24]. The former method selects the optimal weight to maximize the difference between the foreground target and its surrounding background. Besides, the latter method combines individual likelihood maps using equal weights to create a fused version. Li et al. [40] introduce a band attention block, which utilizes the auto-encoder technology for learning the nonlinear spectral relationships. A visual representation of their band attention module can be found in Fig. 7.

To improve the robustness of object characterization features, Li et al. [43] combine temporal, spatial and spectral features in the process of band selection. The improved module is shown in Fig. 8.

However, the limited spectral data has prevented Li et al. from utilizing the semantic features of hyperspectral images. Zhang et al. [93] develop a rapid hyperspectral target tracking method that utilizes a channel selection strategy. It takes into account the variations in both spectral and spatial characteristics within a given frame, and selects three channels,

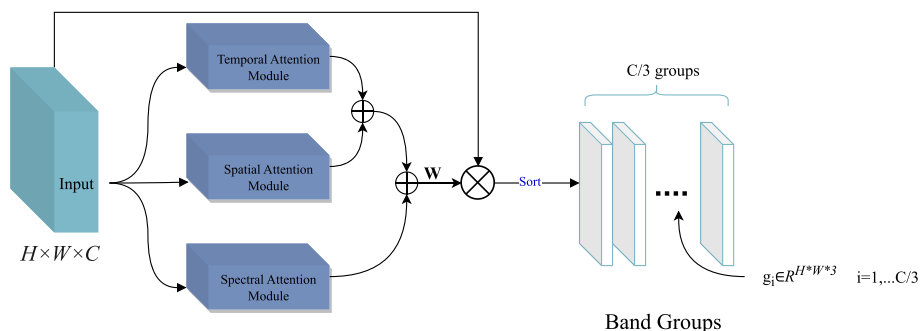


Fig. 8 The improved band attention module in SST-Net [43]

determined by specific modules (entropy, contrast and difference), for increased speed. In our previous work, Wang et al. [80] design a genetic optimization algorithm [53, 90] based on optimum index factor index (or joint entropy) to obtain three bands. A diagram illustrating the process of band selection can be found in Fig. 9.

As shown in Fig. 9, we first divided the initial sequences into several groups via the matrix diagram of the correlation coefficient. Moreover, band selection is completed using an intelligent optimization algorithm. Specifically, the joint information entropy J between two bands, d_i and d_j , in the initial image $D = [d_1; d_2; \dots; d_k]$ with dimensions of $m \times n \times k$, where m to n represent the pixels and k is the number of bands.

$$J(d_i, d_j) = - \sum_{m,n=0}^{255} P_{m,n} \log_2 P_{m,n} \quad (9)$$

where the likelihood of pixel m in band d_i aligning with pixel n in band d_j is represented by the joint probability, $P_{m,n}$. In addition, the crossover, selection and mutation are performed to determine the appropriate bands, iteratively.

To ensure optimal feature extraction, a deep network is utilized and three spectrums are selected through correlation analysis as inputs. Furthermore, Spectral-Spatial-Aware Transformer Fusion Network (SSATFN) [78] mainly uses Hyperspectral Transformer Self-Attention (HTSA) and Hyperspectral Transformer Cross-Attention (HTCA) for fusing multi-band features.

The above data dimensionality reduction methods are employed to enhance the effectiveness of subsequent feature extraction, as hyperspectral video data, despite containing abundant semantic information, exhibits high inter-band correlation, leading to processing redundancy.

4.2 Feature extraction

Feature extraction represents a fundamental concept in the fields of computer vision and image processing, with the principal objective of extracting the most pertinent, informative,

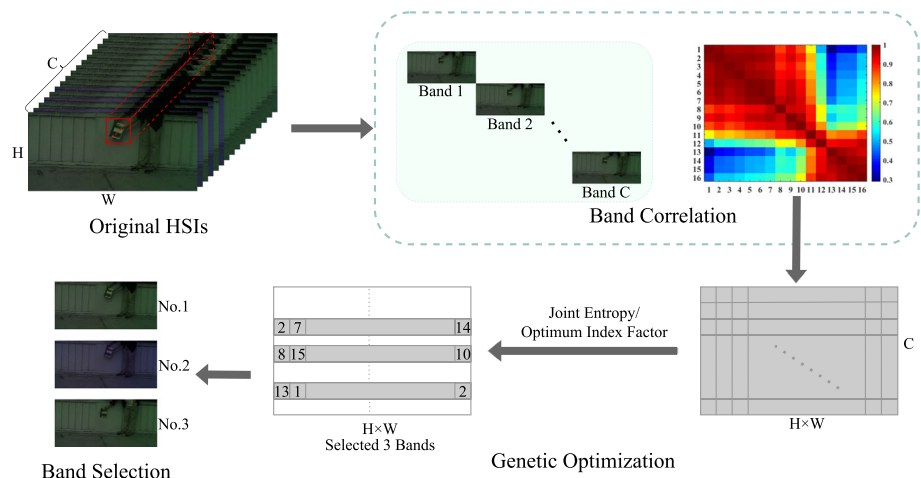


Fig. 9 Band selection in the method [80] proposed by Wang et al. for the initial book images

and discriminative features from raw image data to enable subsequent image processing, analysis, and recognition tasks. It transforms the original data into a compact representation with fewer dimensions, which needs to describe new feature attributes. Nguyen et al. [73] suggest a technique that merges the use of spectral reflectance with a mean shift algorithm for efficient target tracking, but it costs much calculation. In 2009, Banerjee et al. [3] pioneered the use of a Spectral Angle Mapper (SAM) in a hyperspectral video tracking system. Then, Uz Kent et al. [67] propose an innovative sensor capable of dynamic spectral data acquisition by utilizing SAM to evaluate the similarity of spectral vectors in an adaptive multi-mode performance-driven sensor.

In order to obtain a good feature, Qian et al. [54] select the normalized 3D patches as fixed convolutional kernels from the target region to extract features in succeeding frames. Yet, features are not trained, which makes the tracker drift to the background easily. Tu et al. [66] combine local binary pattern features with a sparse representation classifier, improving the classification of hyperspectral processing. Xiong et al. [85] build a three-dimensional local Spectral-Spatial Histogram of Multi-dimensional Gradient (SSHMG) descriptor for HSVs, with the ability to leverage local material information while remaining relatively insensitive to illumination conditions. In addition, the Material Hyperspectral Tracking (MHT) of Xiong makes use of the spatial distribution of materials, which are encoded by abundances and make recognition-related structures discovered. As depicted in Fig. 10, SSHMG and abundances are worked out and then weighted based on their reliability before training the Background-Aware Correlation Filter (BACF). Chen's approach [9] leverages fast spatial-spectral features and achieved real-time updates of the spatial-spectral convolution kernel without the need for offline training. Hou et al. [18] find the last dimension of HOG features contains no information, thus the truncated HOG feature can be introduced in the stacked HOG. Besides, tensor processing is introduced to reduce the differences in the spectral inhomogeneous background, in order to obtain robust and suitable information. Wang et al. [64] also proposed a method called SSTFT where the feature extraction module utilizes a dual-branch approach to extract spectral-spatial features at both coarse-grained and fine-grained levels, subsequently combining them using adaptive weights.

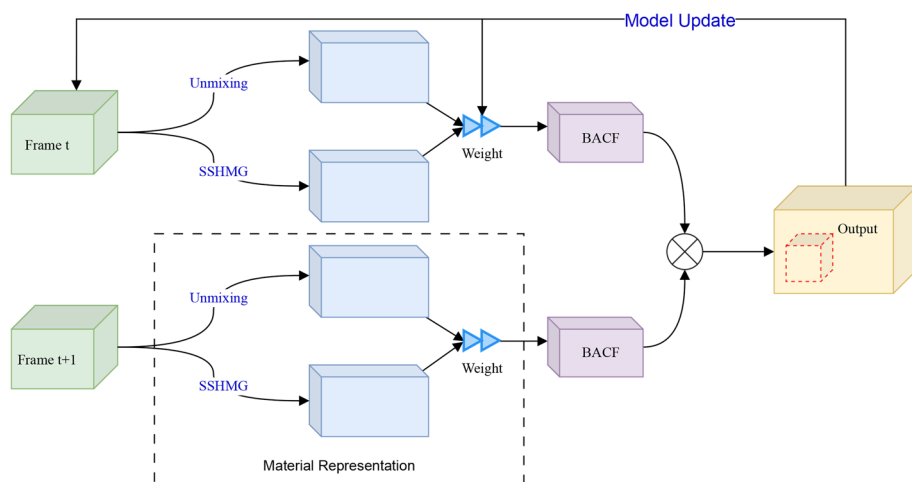


Fig. 10 The structure of MHT [85]

The above methods take full advantage of handcrafted features. In contrast, deep feature extraction has also gained much attention in HOT. Uz Kent et al. [70] propose a Deep Kernelized Correlation Filter-based method (DeepHKCF) without considering valuable spectral information. While tracking aerial objects, HOG feature channels and pure hyperspectral channels are concatenated. A false-color image is generated from the Hyperspectral image before being transferred to the existing AlexNet model. Zhang et al. [95] propose the MFI-HVT (Multi-Features Integration based Hyperspectral Videos Tracker), applying HOG and VGG-19 network [59] for tracking objects. The MFI-HVT also adopts dimension reduction by combining maximum difference selection and the principal component analysis [49, 89]. Liu et al. [37] exploits ResNet-50 [38] as the backbone network, and then to output a response map. Lei et al. [33] adopt a weighted spectral fusion based on structure tensor, considering the contribution of each band to make sure their network is efficient with high performance. The Transformer-based Three-branch Siamese Network (TrTSN) [57] method uses a three-branch Siamese structure to extract the semantic information, and adopts two Transformer-based Fusion Modules (TFM) for generating robust features. The framework of TrTSN is in Fig. 11.

The SSATFN [78] enhances the template branch by applying a multi-scale spectral auxiliary branch, which divides the hyperspectral images into four parts. The DC-HVT [91] divides the feature extractor into two parts, including a depth feature branch where PCA is used to reduce dimensionality, and then the results are input into a ResNet-50 network. Zhang et al. [100] use cube-level structures at multiple scales to extract deep features by applying spectral matching reduction and adaptive-scale three-dimensional hyperspectral features.

The multimode fusion between hyperspectral and RGB videos can also be applied into the field of HOT. Three major challenges, existing in the fusion tasks between the two types of data, include acquiring modality-specific representations, combining multi-modality information, and evaluating hyperspectral and RGB modalities' reliability. Uz Kent et al. [69] use wide field of view color images to eliminate global camera motion, while the narrow field of view hyperspectral images to detect the target of interest by using Supported Vector Machine (SVM). Zhao et al. [92] introduce a classification response aggregation module to effectively combine information from multiple modalities, such as HSV and RGB data. This method helps to mitigate the impact of low-reliability data, and reduce noise.

To sum up, feature extraction referring to the spectral information is important in HOT.

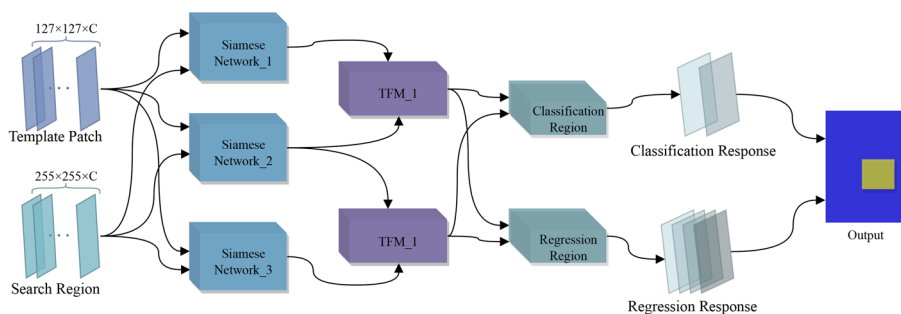


Fig. 11 The structure of TrTSN [57]. TFM—* represents the fusion module and includes the classification prediction and the regression prediction head

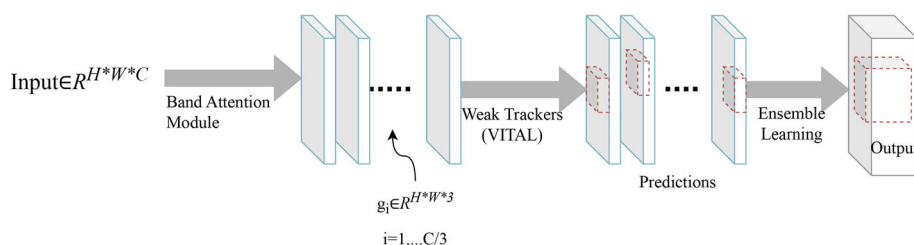


Fig. 12 The structure of BAE-Net [40]

4.3 Tracking frameworks

Up to now, there are two main HOT methods, including CF based method and DL based methods. The DCF [17, 23, 99], with good performance and fast speed, can be utilized to detect the target of the maximum response. Qian et al. [54] firstly combine the CNN with CF for HOT, but it takes fixed positive samples into account and the feature is sensitive to complex background. Xiong et al. [85] apply the material characteristics into the BACF framework for the first time. Yet, the priori spectrum library might be insufficient for another dataset, which is because the abundance features from hyperspectral unmixing mentioned in Section 4.2 are collected offline. In addition, Uz Kent et al. [70] introduce multiple Regions of Interest (ROIs) into the KCF. Liu et al. [45] input fused spectral-spatial features obtained by the deep network into DCF, and the principle of cycle consistency is applied to train the whole network called H^3 Net. Zhang et al. [95] design a tracking approach where each feature generates a weak response map for the object in KCF. These maps are then combined to obtain the final response map through the calculation of adaptive weights. However, the MFI-HVT [95] does not consider scale estimation strategy, leading to tracking drift when facing deformation. For suppressing unexpected peaks of response maps, Hou et al. [18] utilize sparse regularization terms and context-aware information for DCF.

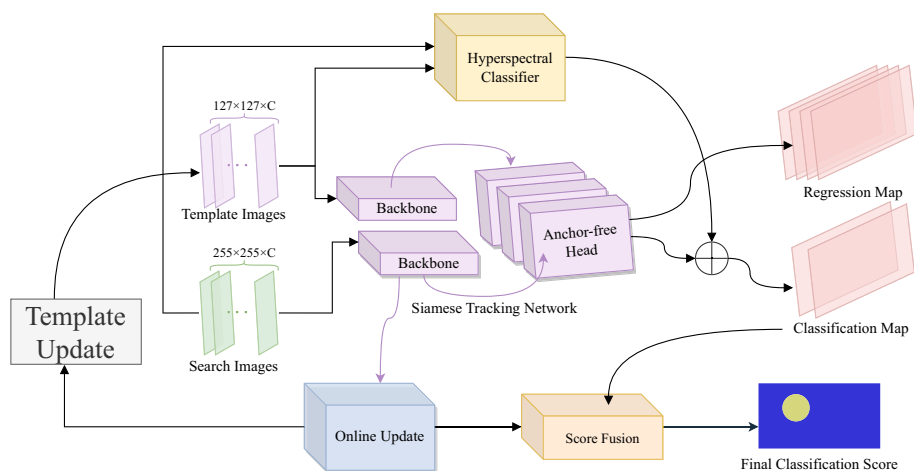


Fig. 13 The structure of HA-Net [37]

There are some workers focusing on deep learning for tracking. Li et al. [40] propose an Attention-Aware Ensemble Network (BAE-Net), which uses the RGB video for network training, which is shown in Fig. 12.

With a spatial-temporal attention module, Li et al. [43] propose a Spectral-Spatial-Temporal attention neural Network (SST-Net). Liu et al. [37] propose an anchor-free siamese network (called HA-Net) which contains RGB classification, regression network and hyperspectral network. As shown in Fig. 13, template images and search images are input into the hyperspectral classification model for learning more information about the difference between foreground and background and a Siamese-based network was used for better knowledge to track.

Additionally, Liu et al. [38] propose a dual siamese network framework (called SiamHYPER) that combines an RGB tracker for pre-training, and a hyperspectral target-aware module through a spatial-spectral cross-attention module. Besides, Lei et al. [33] propose a dual-transfer method called SSDT-Net based on TL and Siamese network, which takes attention to color video information. In our previous algorithm, BS-SiamRPN [80], the genetic algorithm [90] is utilized to reduce the complexity of tracking while preserving most of the image information. Benefiting from TL, more semantic information for tracking can be acquired via the SiamRPN framework. The network is pre-trained using RGB data from the Generic Object Tracking (GOT) [20] benchmark. For solving the problem of little hyperspectral data, the TL strategy is applied to share partial parameters. To be brief, the structure of BS-SiamRPN can be seen in Fig. 14. AD-SiamRPN [63] also used TL for better performance. To counter template information decay in the SiamRPN model, an update network is trained on a general object tracking dataset to accumulate effective templates. Using cumulative templates with spectral data facilitates tracking deformed targets. SiamBAG [61], an innovative Siamese framework for hyperspectral object tracking, minimizes repetition through band attention grouping. Utilizing diverse color tracking data, it employs a classification, regression, and scale-tuning branch to estimate, refine, and adjust object positions, marking a significant advancement in precision and efficiency.

Although some tracking algorithms can balance the speed and the accuracy, such as SiamRPN++ [15], CFNet [13], and DaSiamRPN [98], there are some problems still needing to be dealt with. Firstly, most Siamese tracking algorithms can only distinguish the foreground of the target from the non-semantic background. When the background is cluttered and exists some distractors, the performance of tracking cannot be guaranteed. Secondly, most Siamese trackers need an effective online update model. It is easy to drift to the background because of the update error. In this case, the appearance of the target may change in a very impressive manner. Thirdly, the vulnerability of the local search strategy in the Siamese tracker to partial occlusion of the target is a significant issue. Therefore, some workers are focusing

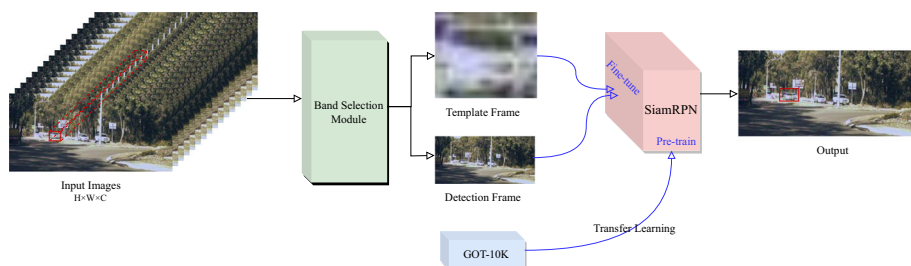


Fig. 14 The flowchart of BS-SiamRPN

on Transformer [60, 62]. Zhao et al. [92] construct a dual branch based on the Transformer Tracking (TransT) structure called RANet. With the attention module, which integrates the features of the template patch and the search area, RANet can maximize the aggregation modules based on HSVs and RGB videos. TrTSN [57] use a classification and regression head based on transformer modules to generate a set of responses, and then merge these responses as the final maps to predict the location of the target, as shown in Fig. 11.

4.4 Model update

Model update is an essential constituent of object tracking, since its objective is to uphold the accuracy and robustness of the model against a wide range of environmental factors and changes in object appearance. The utilization of model update yields several enhancements, such as adaptation to appearance variations, compensation for tracking inaccuracies, handling of occlusions and reappearance, and optimization of tracking performance. Xiong et al. [85] use (10) to compute the group reliability.

$$x^l = (1 - \eta)x^{l-1} + \eta x^l \quad (10)$$

where x denotes a parameter in the model and η is the learning rate. Figure 10 shows MHT's framework, including its model update.

Liu et al. [37] fuse the original template and the updated template, which is obtained by selecting the image with the highest confidence every ten frames to realize an online update, which can be seen in Fig. 13. In order to improve the processing speed, Liu et al. [38] also introduce a spatial-spectral cross-attention mechanism (called SSATTN), which is a module that focuses on individual channels of the input, known as channel-wise attention. SSATTN also enhances the interdependence between the feature extracted by RGB trackers and the channel feature obtained by the hyperspectral target aware module. The SSATFN [78] method utilizes an online update method based generative adversarial network which fine-tunes the generator of feature extraction, gains robust regression, and costs less time. In conclusion, the model update enables the tracker to maintain the target's accurate and robust tracking over time and under various environmental conditions.

5 Evaluation methodology

5.1 Success plot and precision plot

The bounding box overlap is one of the evaluation metrics. The overlap score S can be defined as

$$S = \frac{|b_p \cap b_{gt}|}{|b_p \cup b_{gt}|} \quad (11)$$

where b_p and b_{gt} represent the predicted tracking results and the ground-truth, $\cap$ denotes the intersection operation, and $\cup$ is the union operation.

Based on the overlap score, the number of successful frames whose overlap S is larger than the given threshold is obtained. Then, the success plot [85] shows the ratios of successful frames with the thresholds varying from zero to one. Besides, the Area Under Curve (AUC) of each success plot is utilized to rank the tracking algorithms.

The precision plot [85] is another evaluation metric, designed to assess the accuracy of tracking algorithms. It provides a visual representation of the performance of a tracker by

plotting the percentage of frames, where the center of the estimated bounding box is within a certain distance from the center of the ground truth bounding box. Generally, the score with the threshold of 20 pixels represents the representative precision score for each tracker. The precision plot offers a simple and straightforward comparison of different tracking algorithms, making them a popular choice among researchers. The plot is essential in the development and advancement of object tracking as it highlights the strengths and weaknesses of various methods, enabling researchers to make informed decisions when selecting or designing tracking algorithms and to identify areas for improvement.

5.2 Robustness evaluation

The robustness of one tracker can be impacted and its overall performance might not be accurately represented by using a single starting frame, which is called One-Pass Evaluation (OPE). To address this issue, multiple starting frames can be used to initialize the tracking process and the average performance can be calculated. Two approaches have been proposed to gauge a tracker's ability to handle differing initial conditions, including spatial perturbation of the starting point by starting on various regions of the ground truth (referred to Spatial Robustness Evaluation, SRE) or temporal perturbation through starting at different frames (referred to as Temporal Robustness Evaluation, TRE) [77].

5.3 FPS

FPS (Frames Per Second) [93] is a term commonly used in imaging to indicate the frequency of frame transmission in a given period. The FPS value of a tracking system should be larger than 25. However, it requires special attention for trackers based on deep features.

6 Experimental analysis

To evaluate the performance of the tracker more accurately, we compared the performance of several state-of-the-art algorithms, including MHT [85], DeepHKCF [70], MFI-HVT [95], BAE-Net [40], the SiamRPN method [42] on false-color data, the BS-SiamRPN with TL and without TL, SiamHYPER [38], SiamBAG [61] and AD-SiamRPN [63]. Figures 15–18 below show some qualitative results of these trackers. Some details in experimental configuration can be found in related papers.

In the BS-SiamRPN method, the GOT-10K [20] dataset is utilized for pre-training and 35 hyperspectral testing videos for fine-tuning. The BS-SiamRPN employs the Stochastic Gradient Descent (SGD) [6] method is employed to pre-train the network, and achieved 70 fps on a PC equipped with Nvidia GPU (3080) and 32G RAM.

6.1 Qualitative comparison

The performance of comparison algorithms is processed using four videos in Xiong's dataset, including the *Basketball*, *Forest2*, *Rider2*, and *Student* sequences. Figure 15 shows that the object in video *basketball* has low spatial resolution and moves fast, which makes tracking a difficult problem. At frame #40, all methods can track the target roughly. However, after frame #120, most methods fail to follow the target, because of the rapid movement of the

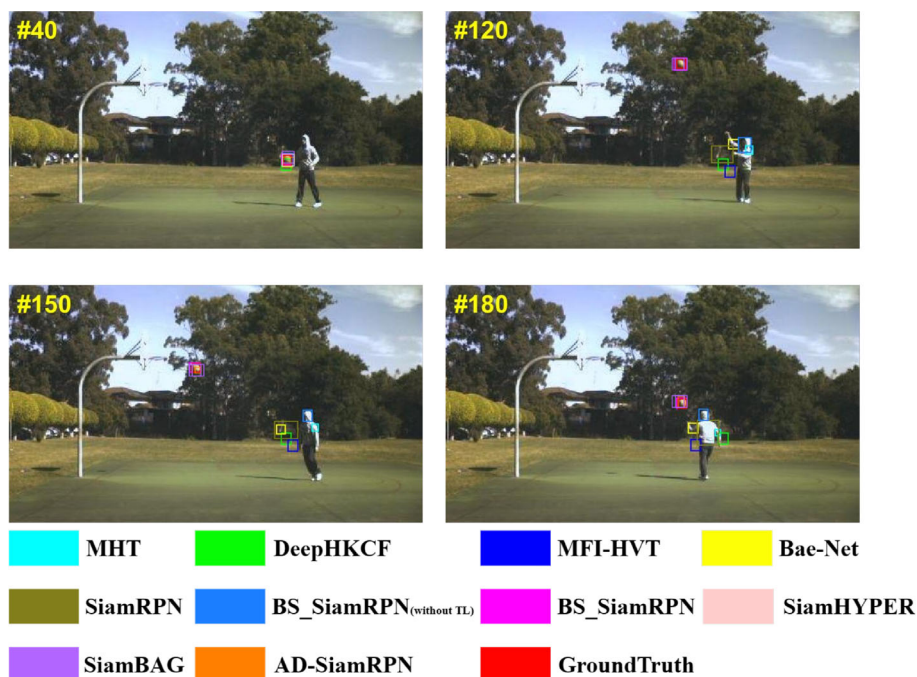


Fig. 15 Qualitative results on the *basketball* sequence

target. They cannot detect the target again until the target returns to the original lost location. By contrast, the Siamese-based methods, including BS-SiamRPN, SiamHYPER, SiamBAG and AD-SiamRPN, successfully track the ball target.

In Fig. 16, the images in *forest2* contain much disturbing semantic information in the background. In fact, the DeepHKCF drifts to the background after frame #50, which is because of the similar color between the clothes and the tree. The two BS-SiamRPN based methods lose targets at frame #150, and only the BS-SiamRPN method with TL can detect the target again at frame #300, due to the fact of that the training process on the large dataset. Unfortunately, both the BAE-net and the MFI-HVT can not find the target after frame #150. The MHT method performs well over this video, which is because it takes the material information into consideration. SiamHYPER excels due to its proficient fusion of deep spatial and spectral features through the spatial-spectral cross-attention module, contributing to its impressive performance.

In Fig. 17, the target moves from near to far, its size gets smaller. In the initial stage, most trackers track the target well. Once the target detection is influenced by the car, the BAE-Net, the SiamRPN and BS-SiamRPN without TL set car as the target template, results in tracking failure. The MHT method, the MFI-HVT deals with the situation of deformation well, which can be seen at frame #300. The last three algorithms based on Siamese perform well due to their powerful model and data augmentation operations.

In Fig. 18, images in *student* contain similar colors in the background. In general, all algorithm can obtain the target location, which proves all trackers handles the situation of

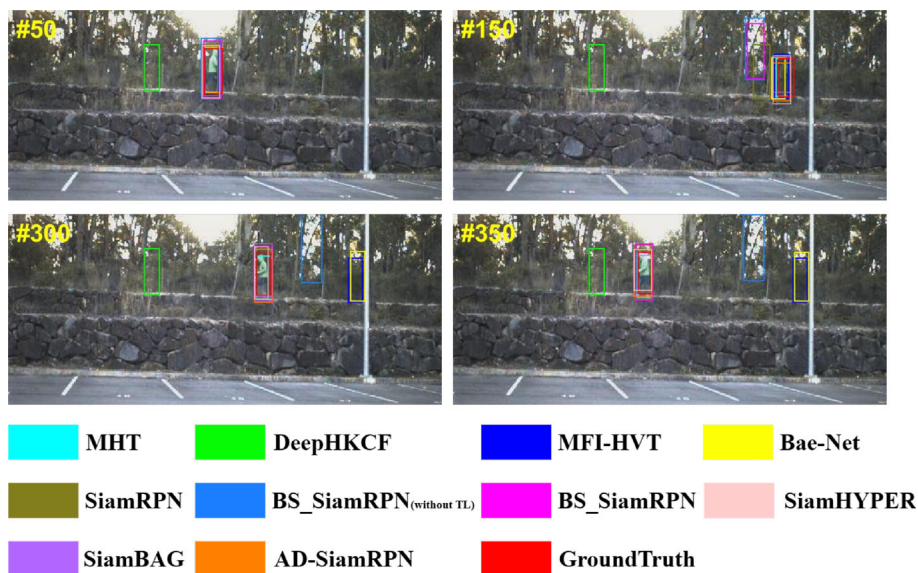


Fig. 16 Qualitative results on the *forest2* sequence

illumination variation well. However, some trackers are sensitive to the target size when the student walks in the hallway. Therefore, the performance of trackers needs to be evaluated using the SRE or TRE.

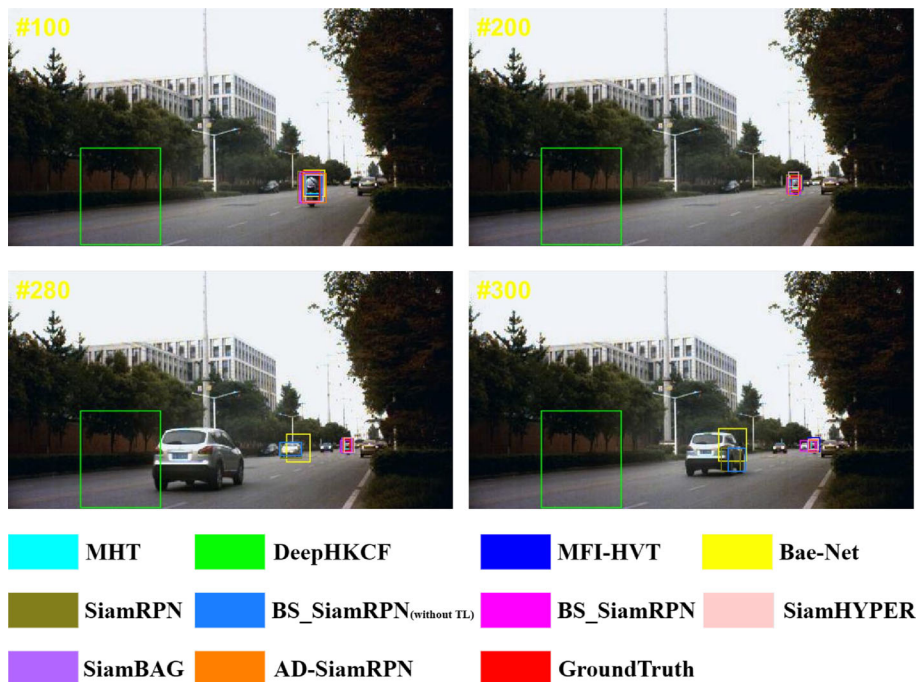


Fig. 17 Qualitative results on the *rider1* sequence

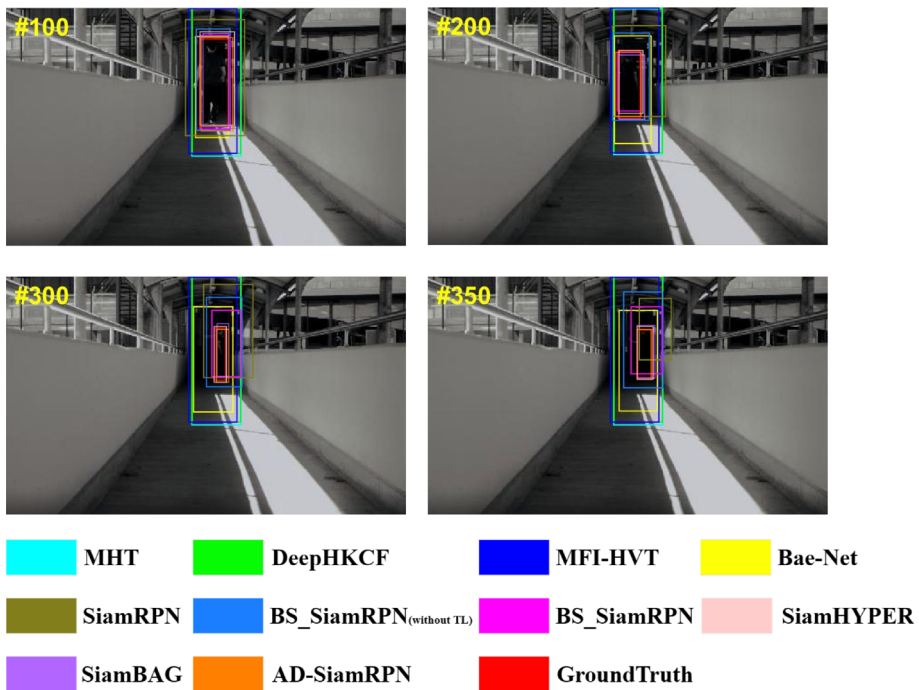


Fig. 18 Qualitative results on the *student* sequence

6.2 Quantitative comparison

In this section, the above algorithms are compared quantitatively with precision and success plots. Table 4 shows the Area Under the Curve (AUC) value of hyperspectral trackers and several RGB trackers. Besides, Fig. 19 illustrates the quantitative performance of these HSV trackers.

The initial trio of algorithms are founded upon Siamese architectures, including SiamHYPER, SiamBAG and AD-SiamRPN, and manifest commendable efficacy, thereby underscor-

Table 4 AUC values of success plots

Algorithm	Video type	AUC	DP@20P
SiamHYPER	HSV	0.678	0.945
SiamBAG	HSV	0.641	0.904
AD-SiamRPN	HSV	0.575	0.861
MFI-HVT	HSV	0.6041	0.8478
MHT	HSV	0.5844	0.8447
DeepHKCF	false-color	0.3861	0.6760
SiamRPN	false-color	0.483	0.7639
BAE-Net	HSV	0.606	0.8764
BS-SiamRPN(Without TL)	HSV	0.468	0.7712
BS-SiamRPN	HSV	0.526	0.8332

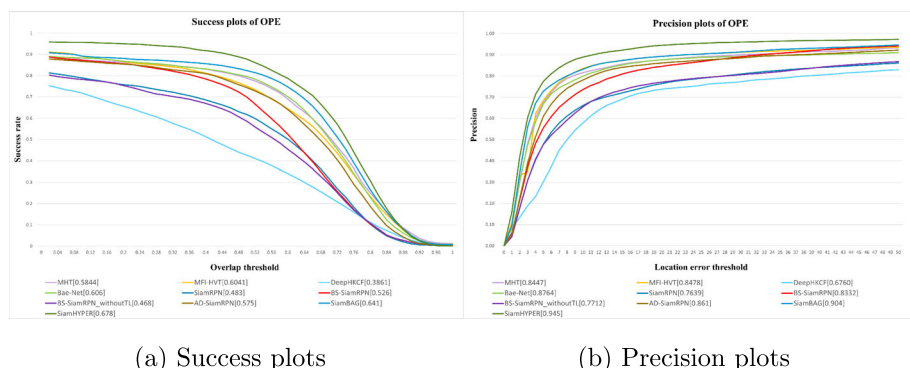


Fig. 19 Success plots and Precision plots of existing algorithms

ing the merits of the Siamese-based model in the tasks of object tracking. Significantly, SiamHYPER emerged as the top-performing algorithm, attaining commendable AUC scores in both precision (0.945) and success (0.678) plots. The BAE-net achieves a good AUC in the precision plot (0.876) and the success plot (0.606). Therefore, spectrum information is effective in the field of HOT. Accordingly, the DeepHKCF method obtains good performance, due to the fact that it does not make full use of spectral information. The same phenomenon appears in the the BS-SiamRPN without TL method. However, with TL using HSVs, the BS-SiamRPN achieves a good AUC score in both plots (0.8332 in the precision plot and 0.526 in the success plot), as shown in Fig. 19.

In short, the hyperspectral information is helpful to improve the classification performance of features. Moreover, the HOT tracker is effective in solving the situation of color similarity between the target and its neighborhood.

7 Conclusions

In this paper, a survey of recent works on hyperspectral video tracking is given, which helps other people better understand hyperspectral trackers. Specifically, experimental results of some popular hyperspectral algorithms, including our previous work, show that hyperspectral information performs better than RGB trackers in accurately separating the target from the background with a similar color. Besides, the feature extraction module in the BAE-Net algorithm is effective in extracting hyperspectral information. In addition, a better model can be optimized via fine-tuning on a recently visible GOT database. With transfer learning, the semantic information referring to HSIs can improve the performance of trackers. To sum up, the survey in this paper lays the foundation for following target recognition.

In the future, the performance of HOT methods can be improved in the following areas.

1. The Siamese network model can be utilized to ensure the speed and accuracy of the HSV tracking.
2. Feature extraction should be taken much attention, no longer just band selection, but rather methods on the original spectral information.
3. With the public NIR hyperspectral videos and several RGB large data sets, transform learning should be of great concern. Then, more works referring to the HOT task will be coming out to increase the ability of target recognition.

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Declarations

Conflicts of interest The authors declare that they have no conflict of interest.

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